

MY1016Z-FAMILY / 250 W DC GEAR MOTOR

Actual sample: KW1016Z-250W24V | ONE COMMON HAND / QUANTITY 2

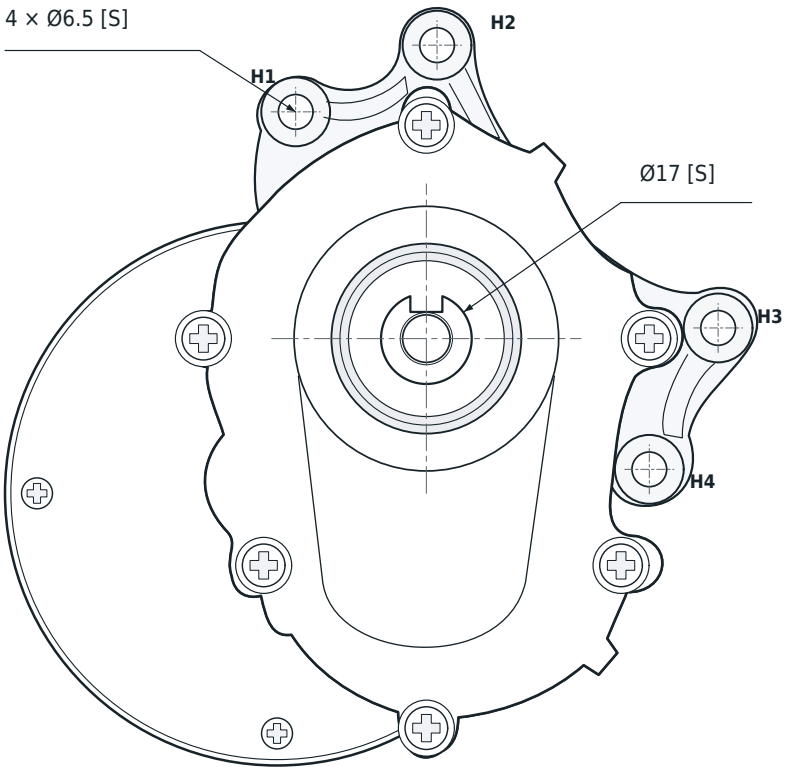
01 GENERAL ARRANGEMENT / RECONSTRUCTED EXTERNAL FORM

SC-MTR-1016Z-REF

REV B / 2026-10-01

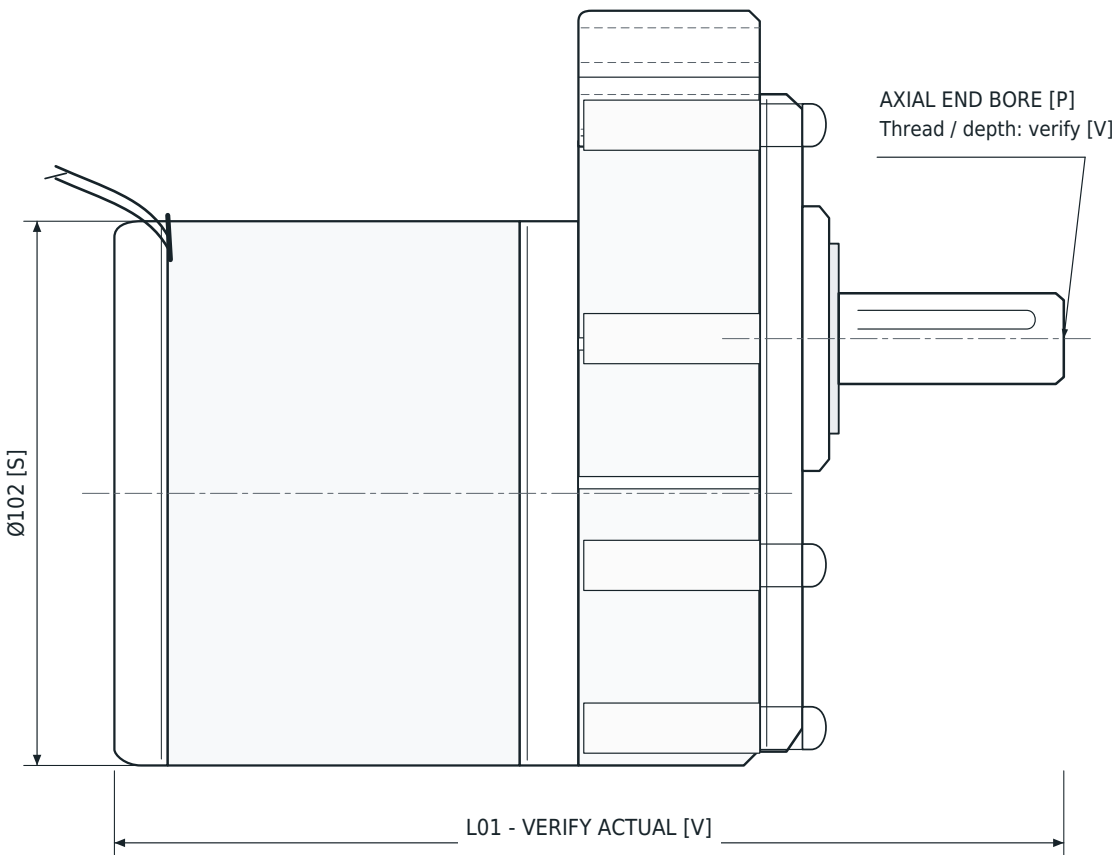
A3 LANDSCAPE / ENGLISH

OUTPUT END / SOURCE CLOCKING



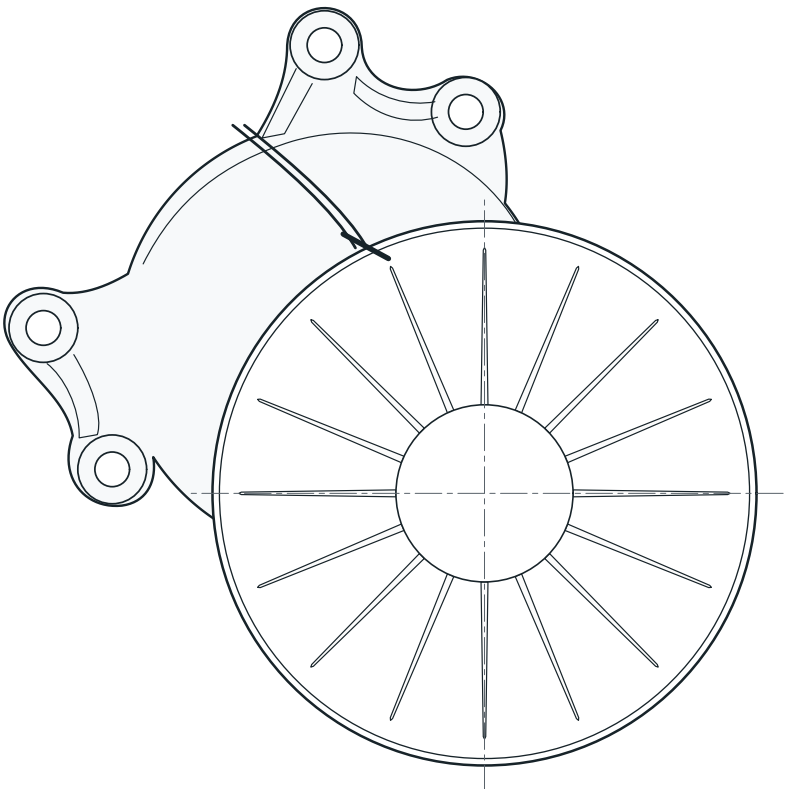
Four mounting bores and six cover screws are distinct features. Bore IDs match Sheet 02.

SIDE VIEW / PHOTO-RECONSTRUCTED



No external threaded spigot or rear brake extension is added to the pictured sample.

REAR / SAME PART, OPPOSITE VIEW



Tapered rear-cap ribs and lead exit follow the photographs. Cable routing is illustrative.

REV B: casting contour, ear webs, cover relief, stepped housing, shaft nose and ribbed rear cap rebuilt from [P] + [S].

Non-dimensioned shape and axial proportions are visual reconstructions, not specimen measurements. All three views: NTS.

ACTUAL-SAMPLE DATA	VALUE / EVIDENCE
Model / motor type	KW1016Z-250W24V / permanent-magnet DC [P]
Rated voltage / output	24 V DC / 250 W output marking [P]
Rated current / speed	13.8 A / 120 r/min (rated, nameplate) [P]
Insulation / enclosure	Class E / IP44 [P]
Electrical input, calculated	24 × 13.8 = 331.2 W at the stated V/I [D]
Unverified operating data	Torque, gear ratio, duty, mass and stall current [V]

INTERFACE SUMMARY	APPLICATION
Mounting / shaft	4 × Ø6.5; Ø17 shaft. Source details: Sheets 02 / 03 [S].
Sample shaft end	Axial end bore [P]; thread, depth and keyway length [V].
Body / overall envelope	Ø102 is source-only [S]; all actual lengths need checking.
Hand / quantity	Two same-handed motors; no mirrored motor defined.
Change scope	Sheet 01 redrawn. Sheets 02-05 retain Rev A content.
Release condition	Confirm both specimens before bracket / hub machining.

[P] PHOTO / NAMEPLATE [S] SUPPLIED SOURCE VARIANT [D] DERIVED [V] VERIFY ACTUAL

Units: mm. No general tolerance is assigned. Do not scale NTS views or photographs.

REFERENCE / VERIFICATION REQUIRED - NOT RELEASED FOR MACHINING

No specimen caliper readings were supplied. Source values do not certify the purchased motor.

Geometry evidence: user photographs [P1] and supplied partial drawing [S1]; register on Sheet 04.

GENERAL ARRANGEMENT - REBUILT

Single-hand motor reference / Smart-Cart Capstone

DRAWING SC-MTR-1016Z-REF	REVISION B	FORMAT A3	SHEET 1 / 5
VIEW SCALE: NTS	mm	EN	QTY 2

MY1016Z-FAMILY / 250 W DC GEAR MOTOR

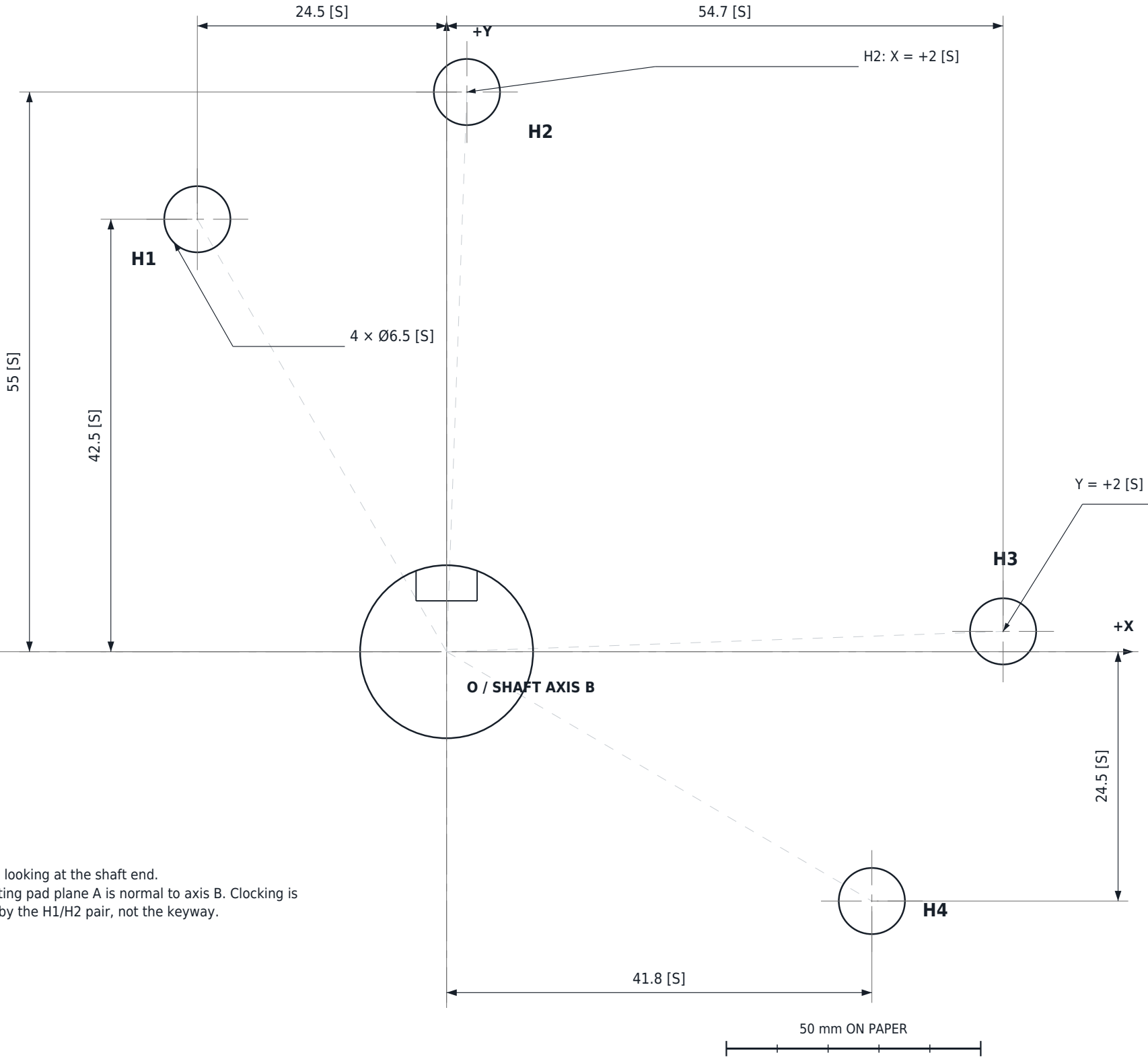
Actual sample: KW1016Z-250W24V | ONE COMMON HAND / QUANTITY 2

02 MOUNTING INTERFACE / SOURCE NOMINAL GEOMETRY

SC-MTR-1016Z-REF

REV A / 2026-10-01

A3 LANDSCAPE / ENGLISH



SOURCE NOMINAL HOLE COORDINATES

HOLE	X	Y	BORE
H1	-24.5	+42.5	Ø6.5 [S]
H2	+2.0	+55.0	Ø6.5 [S]
H3	+54.7	+2.0	Ø6.5 [S]
H4	+41.8	-24.5	Ø6.5 [S]

Coordinates transcribe the supplied drawing [S]; origin and signs are defined above. H4 Y is referenced to the shaft horizontal centerline, not H3.

INDEPENDENT CENTER-DISTANCE CHECKS [D]

PAIR	DISTANCE	STATUS
H1 - H2	29.30	REF only
H2 - H3	74.74	REF only
H3 - H4	29.47	REF only
H4 - H1	94.26	REF only
H1 - H3	88.95	REF only
H2 - H4	88.91	REF only

Calculated from the nominal coordinates: $c = \sqrt{(\Delta X^2 + \Delta Y^2)}$. These are redundant checking dimensions, not new toleranced requirements.

HOLD POINT - BRACKET INTERFACE

Confirm all four bores, six center distances, pad coplanarity and clocking on both actual motors. Cover screws are not mounting bores. Ø6.5 is a bore diameter, not an M6 thread. Do not replace this pattern with the earlier approximate 30 / 73.5 mm values.

[P] PHOTO / NAMEPLATE [S] SUPPLIED SOURCE VARIANT [D] DERIVED [V] VERIFY ACTUAL

Units: mm. No general tolerance is assigned. Do not scale NTS views or photographs.

REFERENCE / VERIFICATION REQUIRED - NOT RELEASED FOR MACHINING

No specimen caliper readings were supplied. [S] values are not measured KW1016Z dimensions.

Both purchased motors use this same part definition; installation rotation is an assembly task.

MOUNTING INTERFACE

Single-hand motor reference / Smart-Cart Capstone

DRAWING SC-MTR-1016Z-REF	REVISION A	FORMAT A3	SHEET 2 / 5
VIEW SCALE: 2:1 (interface)	mm	EN	QTY 2

MY1016Z-FAMILY / 250 W DC GEAR MOTOR

Actual sample: KW1016Z-250W24V | ONE COMMON HAND / QUANTITY 2

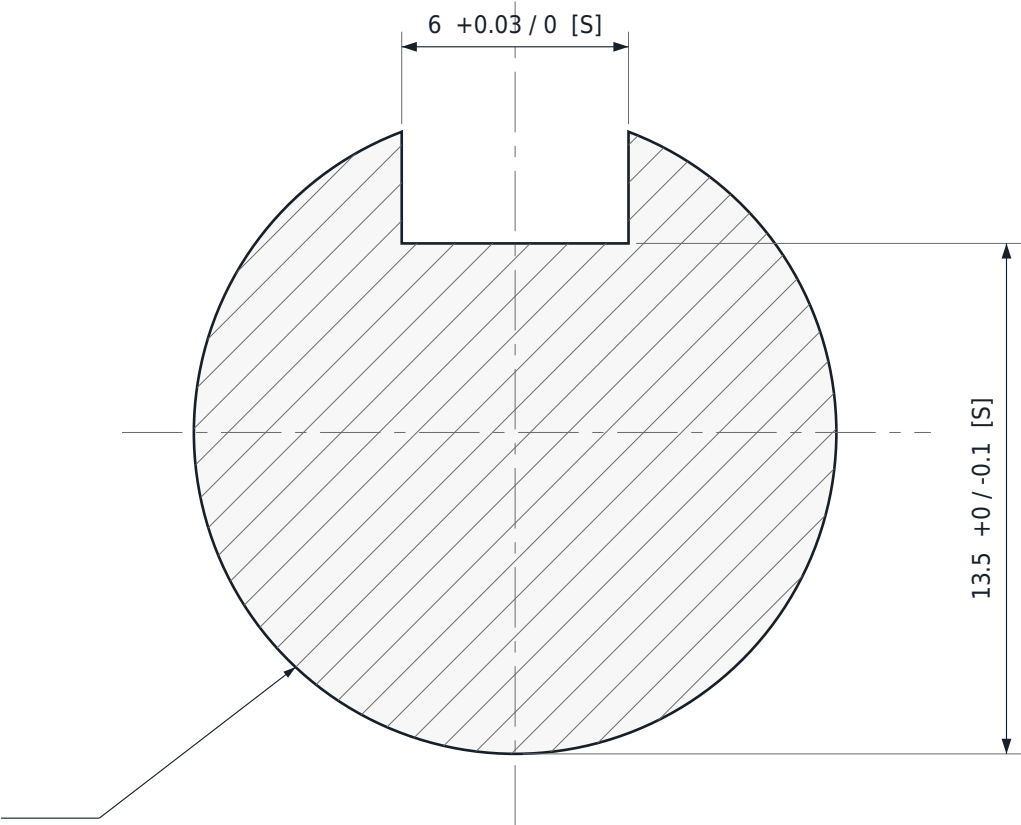
03 OUTPUT SHAFT / KEYWAY / END-FEATURE DETAILS

SC-MTR-1016Z-REF

REV A / 2026-10-01

A3 LANDSCAPE / ENGLISH

DETAIL K / SOURCE SHAFT CROSS-SECTION [S]

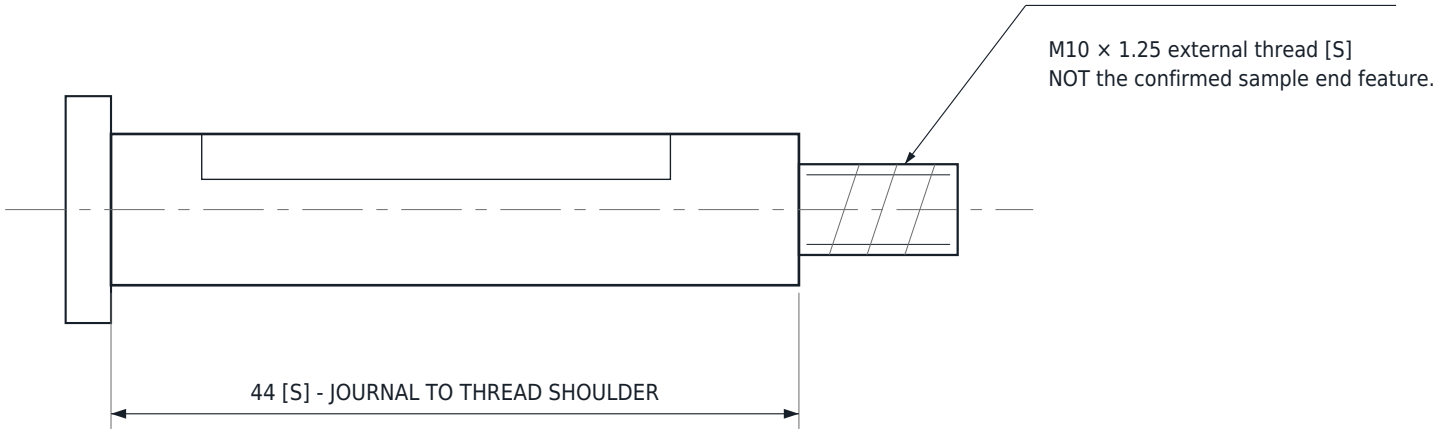


Ø17 -0.006 / -0.024 [S]
Nominal section scale: 5:1 on A3.

SOURCE FEATURE	NOMINAL / DEVIATIONS	DERIVED SOURCE LIMITS
Shaft diameter	Ø17 -0.006/-0.024	16.976 to 16.994
Keyway width	6 +0.03/0	6.000 to 6.030
Bottom-to-floor height	13.5 +0/-0.1	13.400 to 13.500
Nominal keyway depth	17 - 13.5 = 3.5 [D]	REF; not a new tolerance

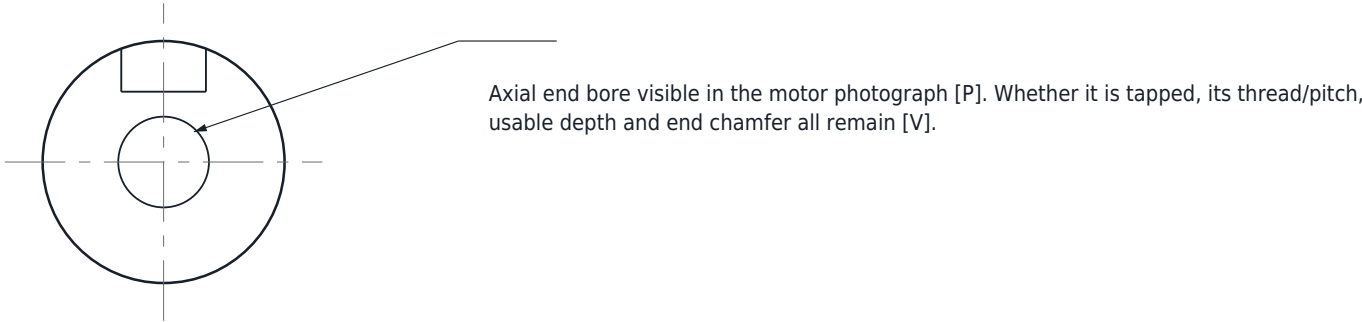
Only source tolerances are transcribed. The shaft diameter band is 0.018 mm wide; verify with an instrument and measurement process adequate for that band, not display resolution alone.

SOURCE SHAFT SIDE VIEW / VARIANT ONLY [S]



The source 44 mm ends at the start of the threaded tip; it is not the total shaft projection. Source thread length and total projection are not dimensioned.

ACTUAL END FEATURE / DO NOT SUBSTITUTE SOURCE THREAD



Keyway width, base height, length and angular position must also be confirmed. The 5:1 detail is a source section, not a certified fit specification for the KW1016Z sample.

[P] PHOTO / NAMEPLATE [S] SUPPLIED SOURCE VARIANT [D] DERIVED [V] VERIFY ACTUAL

Units: mm. No general tolerance is assigned. Do not scale NTS views or photographs.

REFERENCE / VERIFICATION REQUIRED - NOT RELEASED FOR MACHINING

No specimen caliper readings were supplied. [S] values are not measured KW1016Z dimensions.

Both purchased motors use this same part definition; installation rotation is an assembly task.

SHAFT AND KEYWAY

Single-hand motor reference / Smart-Cart Capstone

DRAWING SC-MTR-1016Z-REF	REVISION A	FORMAT A3	SHEET 3 / 5
VIEW SCALE: 5:1 / NTS	mm	EN	QTY 2

MY1016Z-FAMILY / 250 W DC GEAR MOTOR

Actual sample: KW1016Z-250W24V | ONE COMMON HAND / QUANTITY 2

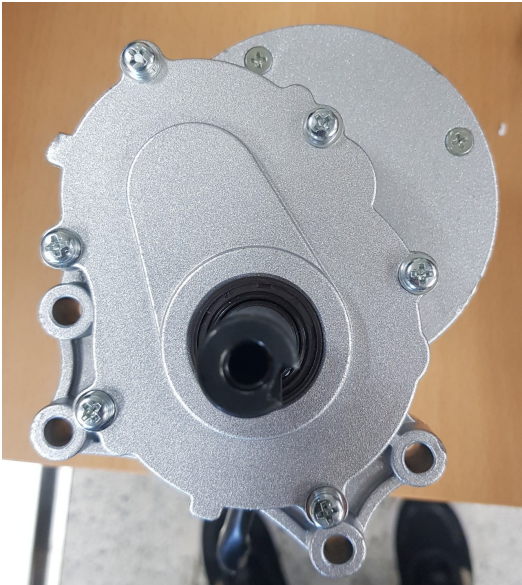
04 SOURCE RECONCILIATION / PHOTO EVIDENCE

SC-MTR-1016Z-REF

REV A / 2026-10-01

A3 LANDSCAPE / ENGLISH

ACTUAL PHOTOGRAPHS / GEOMETRY EVIDENCE [P]



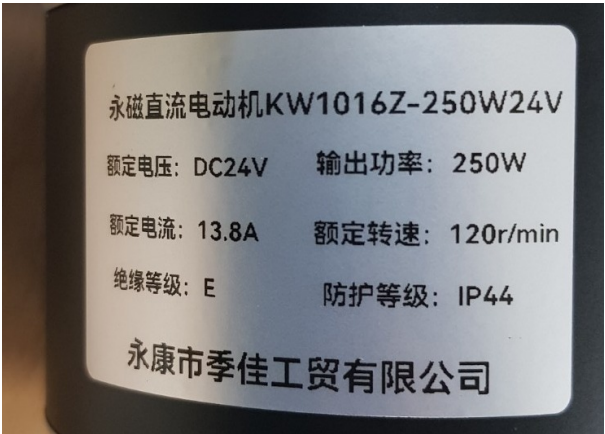
Output end / four mounting ears



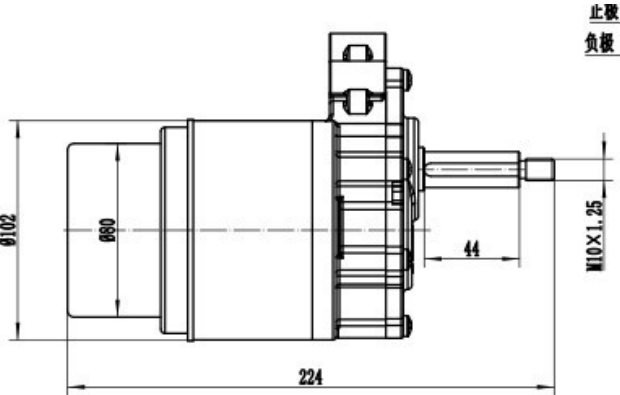
Side / unextended rear housing



Rear / no visible release lever



Nameplate: KW1016Z-250W24V; 24 V; 250 W; 13.8 A; rated 120 r/min; E; IP44.



Supplied source: ZD101AZ-WX-02, L (250W-24V). Rear attachment and external threaded tip are visible; the full 224 mm dimension belongs to that source configuration.

SOURCE-VARIANT APPLICABILITY

FEATURE	ACTUAL SAMPLE [P]	SOURCE ONLY [S]
Identity	KW1016Z-250W24V	ZD101AZ-WX-02; L
Voltage / output	24 V / 250 W	24 V / 250 W
Current	13.8 A, rated	13.4 A, rated
Speed	120 r/min, rated	110 rpm, no load
Motor body Ø	Measure actual [V]	102 [S]
Rear attachment Ø	Not applicable as drawn	80 [S]; rear attachment
Overall axial length	Measure actual [V]	224 [S]; full source unit
End feature	Axial bore visible [P]	M10 × 1.25, external [S]
Journal length	Measure actual [V]	44 [S]; excludes tip
Rear brake / lever	Not visible in photos	Attachment / lever shown
Lead specification	Size / length [V]	2 mm²; 300 exposed [S]
Mounting holes	Verify on both [V]	4 × Ø6.5 [S]

WHAT THIS REVISION DOES NOT CLAIM

The source drawing is not an exact drawing of the purchased sample. Overall 224, rear Ø80, external M10 thread, 110 rpm no-load speed and 13.4 A must not overwrite the physical sample data. No unmeasured envelope or torque is approved here.

EVIDENCE REGISTER

[P1] User motor photographs: 20261001_113605 / 113630 / 113653 / 113711 / 113731.
[S1] User-supplied partial source drawing: image(1).png. English callouts in this PDF are transcriptions.
[R1] Project V8 3D/CAD specification (2026-09-21) / component-register.csv: no actual caliper dataset.

[P] PHOTO / NAMEPLATE [S] SUPPLIED SOURCE VARIANT [D] DERIVED [V] VERIFY ACTUAL

Units: mm. No general tolerance is assigned. Do not scale NTS views or photographs.

REFERENCE / VERIFICATION REQUIRED - NOT RELEASED FOR MACHINING

No specimen caliper readings were supplied. [S] values are not measured KW1016Z dimensions.

Both purchased motors use this same part definition; installation rotation is an assembly task.

EVIDENCE AND APPLICABILITY

Single-hand motor reference / Smart-Cart Capstone

DRAWING SC-MTR-1016Z-REF	REVISION A	FORMAT A3	SHEET 4 / 5
VIEW SCALE: NTS	mm	EN	QTY 2

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05 PHYSICAL VERIFICATION / TWO IDENTICAL SPECIMENS

SC-MTR-1016Z-REF

REV A / 2026-10-01

A3 LANDSCAPE / ENGLISH

Purpose: complete one verified part record for two same-handed specimens. Use the coordinate system on Sheet 02. Record actual units and instrument identity; a blank cell is not an accepted dimension.

SPECIMEN A ID: _____ SPECIMEN B ID: _____ INSTRUMENT / SERIAL / ACCURACY: _____

ID	ACTUAL FEATURE TO RECORD	REFERENCE, NOT ACCEPTANCE	SPECIMEN A	SPECIMEN B	METHOD / INTERFACE NOTE
D01	Motor cylinder outside diameter	102 [S]			Outside measurement
L01	Complete axial length, back to shaft tip	Actual [V]; not 224			Exclude cable; include end features
L02	Mounting contact plane A to shaft tip	Actual [V]			Pad plane must be identified
L03	Motor can length / rear cap length	Actual [V]			Record lengths separately
L04	Gearcase axial thickness / front boss projection	Actual [V]			No source axial dimensions
E01	Maximum transverse width × height	Actual [V]			Include ears and cover screw heads
E02	Can-axis offset from output axis (X, Y)	Actual [V]			Record signed components
D02	Output shaft diameter	17 [S]; limits on 03			Measure at several axial / angular sites
K01	Keyway width / bottom-to-floor height	6 / 13.5 [S]			Check source tolerances separately
K02	Keyway axial length / angle to X-axis	Actual [V]			Angle is not defined by source section
T01	End bore: plain / tapped; diameter or thread	Actual [V]			Do not assume source M10 × 1.25
T02	Bore usable depth / thread engagement	Actual [V]			Use depth and thread gauges as needed
H01	Mounting bore diameters H1 / H2 / H3 / H4	4 × 6.5 [S]			Four bores; not the cover fasteners
H02	H1 coordinate (X, Y)	(-24.5, +42.5) [S]			Use common datums and clocking
H03	H2 coordinate (X, Y)	(+2.0, +55.0) [S]			Do not round X offset to zero
H04	H3 coordinate (X, Y)	(+54.7, +2.0) [S]			Do not round Y offset to zero
H05	H4 coordinate (X, Y)	(+41.8, -24.5) [S]			Y from shaft horizontal axis
H06	Six hole-to-hole center distances	Sheet 02 [D]			Independent closure check
M01	Pad / ear thickness; face coplanarity	Actual [V]			Confirm all four contact surfaces
M02	Mounting-bolt access / cover-head projection	Actual [V]			Include washer and tool keepout
C01	Cable exit; free length; connector and bend space	Actual [V]			Source 2 mm² / 300 is not confirmed
W01	Assembly mass	Actual [V]			Weigh with defined cable inclusion

Measurement consistency: for two equal bores of diameter d, center distance = outer span - d = inner gap + d. For unequal bores, use half the sum of their diameters. Check all six distances; record values rather than forcing them to match the source.

Release criteria: verify both specimens, resolve source/part mismatches, update all [V] entries, check hub retention and bracket clearances, then issue a new revision. This PDF does not authorize changes to the gearbox or safety systems.

RECORDED BY: _____ DATE: _____ REVIEWED BY: _____ NEXT REVISION / STATUS: _____

[P] PHOTO / NAMEPLATE [S] SUPPLIED SOURCE VARIANT [D] DERIVED [V] VERIFY ACTUAL Units: mm. No general tolerance is assigned. Do not scale NTS views or photographs. REFERENCE / VERIFICATION REQUIRED - NOT RELEASED FOR MACHINING No specimen caliper readings were supplied. [S] values are not measured KW1016Z dimensions. Both purchased motors use this same part definition; installation rotation is an assembly task.	MEASUREMENT RECORD			
	Single-hand motor reference / Smart-Cart Capstone			
	DRAWING SC-MTR-1016Z-REF	REVISION A	FORMAT A3	SHEET 5 / 5
	VIEW SCALE: NTS	mm	EN	QTY 2